

tf.fill

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L204-L250\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L204-L250)

Creates a tensor filled with a scalar value.

```
tf.fill(  
    dims, value, name=None, layout=None  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
Ragged tensors (https://www.tensorflow.org/guide/ragged_tensor) tf.data: Build TensorFlow input pipelines (https://www.tensorflow.org/guide/data) Subword tokenizers (https://www.tensorflow.org/text/guide/subwords_tokenizer) Unicode strings (https://www.tensorflow.org/text/guide/unicode)	Scalable model compression (https://www.tensorflow.org/tutorials/optimization/compression) Multiple changepoint detection and Bayesian model selection (https://www.tensorflow.org/probability/examples/Multiple_changepoint_detection_and_Bayesian_model_selection) Graph-based Neural Structured Learning in TFX (https://www.tensorflow.org/tfx/tutorials/tfx/neural_structured_learning) Graph regularization for sentiment classification using synthesized graphs (https://www.tensorflow.org/neural_structured_learning/tutorials/graph_keras_lstm_imdb) Bayesian Gaussian Mixture Model and Hamiltonian MCMC (https://www.tensorflow.org/probability/examples/Bayesian_Gaussian_Mixture_Model)

See also [tf.ones](https://www.tensorflow.org/api_docs/python/tf/ones) (https://www.tensorflow.org/api_docs/python/tf/ones), [tf.zeros](https://www.tensorflow.org/api_docs/python/tf/zeros) (https://www.tensorflow.org/api_docs/python/tf/zeros), [tf.one_hot](https://www.tensorflow.org/api_docs/python/tf/one_hot) (https://www.tensorflow.org/api_docs/python/tf/one_hot), [tf.eye](https://www.tensorflow.org/api_docs/python/tf/eye) (https://www.tensorflow.org/api_docs/python/tf/eye).

This operation creates a tensor of shape `dims` and fills it with `value`.

For example:

```
>>> tf.fill([2, 3], 9)  
<tf.Tensor: shape=(2, 3), dtype=int32, numpy=  
array([[9, 9, 9],  
       [9, 9, 9]], dtype=int32)>
```

[tf.fill](https://www.tensorflow.org/api_docs/python/tf/fill) (https://www.tensorflow.org/api_docs/python/tf/fill) evaluates at graph runtime and supports dynamic shapes based on other runtime `tf.Tensor`s, unlike [tf.constant\(value, shape=dims\)](https://www.tensorflow.org/api_docs/python/tf/constant) (https://www.tensorflow.org/api_docs/python/tf/constant), which embeds the value as a `Const` node.

Args	
<code>dims</code>	A 1-D sequence of non-negative numbers. Represents the shape of the output tf.Tensor (https://www.tensorflow.org/api_docs/python/tf/Tensor). Entries should be of type: <code>int32</code> , <code>int64</code> .
<code>value</code>	A value to fill the returned tf.Tensor (https://www.tensorflow.org/api_docs/python/tf/Tensor).
<code>name</code>	Optional string. The name of the output tf.Tensor (https://www.tensorflow.org/api_docs/python/tf/Tensor).
<code>layout</code>	Optional, tf.experimental.dtensor.Layout (https://www.tensorflow.org/api_docs/python/tf/experimental/dtensor/Layout). If provided, the result is a DTensor (https://www.tensorflow.org/guide/dtensor_overview) with the provided layout.
Returns	
A tf.Tensor (https://www.tensorflow.org/api_docs/python/tf/Tensor) with shape <code>dims</code> and the same dtype as <code>value</code> .	
Raises	
<code>InvalidArgumentError</code>	<code>dims</code> contains negative entries.
<code>NotFoundError</code>	<code>dims</code> contains non-integer entries.

numpy compatibility

Similar to `np.full`. In `numpy`, more parameters are supported. Passing a number argument as the shape (`np.full(5, value)`) is valid in `numpy` for specifying a 1-D shaped result, while TensorFlow does not support this syntax.

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Last updated 2024-04-26 UTC.